K1263 — $\sin^2\theta$_W = 3/13 FINAL: unanimous Structural/Identified. Cal's σ-objection raised and correctly retracted. The tier is UNFORCED-limited, not σ-limited. Linear-algebra cast + promotion gate.

Keeper, 2026-08-07. The crux (K1261) is settled. All four CIs verified it independently and converged, including a real adversarial pass by Cal that he then resolved himself. This closes the $\sin^2\theta$_W re-examination and sets the forward research frame per Casey's "connect to the corpus, linear algebra on D_IV^5."

Final ruling (unanimous — Keeper, Lyra, Elie, Cal)

Three distinct objects, three distinct tiers: 1. $c_5/c_3 = 3/13$ as a Chern number of Q^5: Derived-as-mathematics. Lyra and Cal each expanded $c(TQ^5) = (1+h)^7/(1+2h)$ from scratch → {1,5,11,13,9,3}, no physics input, no foreknowledge of 3 or 13. Target-innocent. Keep it. 2. The identification $\sin^2\theta$_W = c_5/c_3 = 3/13: Structural/Identified. A real structural reading (color / total-gauge-dimension ratio), unforced, MS-bar/M_Z-scheme-caveated. NOT Proved (the identification isn't forced — the over-claim was real), NOT retired (the number is real and the shadow argument is false), NOT Candidate (see the σ adjudication below). 3. The $\|B-L\|^2 = 28/3 \to 3/13$ derivation route: REFUTED, stays dead (K738 + Cal §32, RH charges). Killing that one route never licensed retiring the independent topological reading — that was the over-demotion.

The retirement's stated ground is quantitatively false, now verified three independent ways: 3/8 runs DOWN to ≈0.20–0.208, not 0.2308 — Elie's non-circular minimal-SU(5) form $\sin^2\theta$_W = 1/6 + (5/9)(α/α_s) = 0.203 (1-loop), Cal's 0.208 (2-loop), both independent of the measured $\sin^2$. 3/13 = 0.2308 is a *different number* from where 3/8 runs. 3/13 is not the shadow of 3/8.

Adjudicating Cal's σ-objection (the important part — an auditor resolving an auditor)

Cal (§339) refuted "Structural/Identified," arguing Candidate: scored in σ, 3/13 = 0.23077 is ~11σ from the MS-bar value ($\sin^2\theta$_W measured to ~0.02%), so "0.19% match" is the dev%-trap — 0.19% on a precision constant is a catastrophe, not a hit; and since the identification is unforced, "you score it, and scored it's 11σ."

Cal (§340) retracted this himself after reading both papers. He is right to retract, and the reason is load-bearing for how we tier this class of result:

- The σ-score is a cross-region category error — the region-rule forbids it. 3/13 = c_5/c_3 is a scale-free topological ratio (an interior/discrete object). $\sin^2\theta$_W(M_Z, MS-bar) is a scale- and scheme-specific exterior measurement. Scoring the interior ratio against the exterior value *in* σ is exactly the interior-vs-exterior comparison the standing rule prohibits ([[feedback_region_match_sigma_scoring_interior_vs_exterior]]). The 11σ is not a real refutation; it's the category error firing. Cal's §339 was the region-rule failure he'd flagged in himself before, recurring.
- So the tier is UNFORCED-limited, not σ-limited. The reason 3/13 sits at Structural/Identified and no higher is NOT "it's 11σ off" — it's that the *identification is unforced*: nothing forces c_5/c_3 (rather than c_4/c_2 = 9/11, c_5/c_1 = 3/5, ...) to be the mixing angle, and nothing forces the scale-free ratio to be read at M_Z MS-bar. That unforced double-choice is the whole limitation. The σ does not belong in the tier.
- "Circular" and "unforced" are the same observation. Cal noted the Chern "theorem" is circular — it re-expresses the pre-existing N_c/(N_c+2n_C) as c_5/c_3 because the coefficients happen to be 3 and 13. Correct: the Chern *numbers* are target-innocent, but the *choice of that ratio* is post-hoc (the paper's own Section 5 admits "only at n=5 does the ratio give a clean fraction"). The numbers are real; the ratio-selection is unforced. That is precisely Structural/Identified.

Verdict: Candidate is withdrawn; Structural/Identified stands, unanimously. Cal's objection was valuable — it forced us to name *why* it's Structural (unforced choice) rather than lazily reading "0.19% = a match." The honest reason is the unforced-ness, and the σ is inadmissible here.

What K739 got RIGHT (preserve the true bottom line)

Cal's sharpest point: K739's conclusion — "$\sin^2\theta$_W is NOT a BST precision win" — was correct; only its mechanism (the 3/8-shadow) was false. The honest state of $\sin^2\theta$_W in BST: two routes that *disagree* — - 3/8 (fermion-content, Tr T_3^2/Tr Q^2): forced, but runs to ≈0.208, missing observed 0.231 by ~10% (no better than a non-SUSY GUT). - 3/13 (Chern c_5/c_3): unforced, matches 0.231 at 0.19% but with an unexplained scale/scheme-match.

Neither is a clean win. $\sin^2\theta$_W remains an honest open tension, not a triumph. The over-demotion error was denying the *real Chern number*, NOT the "not a precision win" bottom line. We restore the number to its honest tier; we do NOT restore a "win."

Linear-algebra cast on D_IV^5 (Casey's directive — the forward frame)

The Chern classes c_k(Q^5) are the elementary symmetric polynomials e_k of the Chern roots = the K = SO(5)×SO(2) weights of the tangent space at o = the eigenvalues of the curvature operator on T_o(D_IV^5). So:

$\sin^2\theta$_W (Chern route) = c_5/c_3 = e_5/e_3 — a ratio of two elementary-symmetric invariants of the curvature-eigenvalue set of D_IV^5's tangent bundle.

This is the corpus's "every observable is a matrix element / eigenvalue / grading of one operator" program ([[feedback_linear_algebra_on_div5]]) applied to the Weinberg angle: it is a *grading ratio* of the curvature operator. The tangent decomposition T_o = $\mathbb{C}^3$_color ⊕ $\mathbb{C}^2$_weak (paper Section 3.1) is the isotropy-rep block structure of that same operator under SU(3)×SU(2)×U(1) ⊂ K. So the physical claim "$\sin^2\theta$_W = confined/total gauge dimension" is the claim that the mixing reads the color-block grading of the curvature operator.

The promotion gate (Structural → Derived) — well-defined, and it is the research target

To promote the identification, force the ratio-choice: give a geometric/index-theoretic reason the physical mixing angle equals e_5/e_3 (the color-block grading) specifically — not e_4/e_2 or e_5/e_1, and read at the geometric scale. The seed is the Lefschetz fixed-point reading (paper Section 3.2: c_5 = 3 counts the color-confined tangent fixed points, c_3 = 13 the total gauge-relevant directions). The gate: is that a *forcing* or an interpretation? If forced, it would both derive 3/13 AND explain why the tangent-bundle reading beats the fermion-content trace (3/8) — the real prize. This is a live, honest, well-posed target, appropriately open. Do not bank it; investigate it.

Actions

- Sweep GREEN-LIT as a tier-correction (Lyra runs it): across the ~45 files + 2 dedicated Weinberg papers, "Proved / 0.19% match" → "Structural/Identified: Chern ratio c_5/c_3 proved as topology; $\sin^2\theta$_W identification unforced, MS-bar/M_Z-scheme-caveated." Plus four specific over-claim fixes Cal itemized (beyond the tier header): (1) the "doesn't run / running already encoded in the geometry" passages (paper Sections 6.2–6.3) are physically unsupported — $\sin^2\theta$_W *does* run and 3/8 runs to 0.208; correct them. (2) The Chern "dictionary" (Section 4) is match-cheap, not over-determination — it admits c_2 = 11 has no meaning and Cabibbo is 3.1% off; frame it honestly. Papers reframed, not retracted — they document a real Chern theorem with a properly-caveated physics reading, which is a better outcome than deletion. Leave every legitimate c_3 = 13 use untouched (f_D/f_π, α-binding, the Chern integer itself).
- Registry: T197 + T1919 already corrected to Structural/Identified (K1261). const_011 stays Structural/Identified (Grace corrected the "runner" note).
- Externals: once the sweep reads no "Proved" anywhere and is Keeper-clean, the QM-10/10 block lifts (the flagship does not use $\sin^2\theta$_W; the block was the registry contradicting the honest tier, now resolved).
- θ_{13} demotion: unaffected, stands (genuine competing-forms hygiene).

— Keeper, K1263, 2026-08-07. $\sin^2\theta$_W = 3/13 FINAL, unanimous: Chern ratio = Derived-math; identification = Structural/Identified (unforced ratio-choice + scale/scheme-match); B−L route refuted. Cal's σ→Candidate objection RAISED and correctly RETRACTED (σ is a cross-region category error the region-rule forbids; tier is unforced-limited not σ-limited). K739's "not a precision win" bottom line was RIGHT; its 3/8-shadow mechanism was false (3/8→0.203–0.208≠0.2308, three independent computations). Linear-algebra cast: $\sin^2\theta$_W = e_5/e_3 of the tangent-bundle curvature eigenvalues = a color-block grading ratio. Promotion gate = force the ratio-choice (Lefschetz seed). Sweep green-lit as tier-correction + 4 paper over-claim fixes; papers reframed not retracted. Externals lift when Keeper-clean.